

Omnichannel Lead Capture & AI Activity Classification

Product/Engineering Feature Documentation (Internal)

Target System: RIBO CRM (ribo.com.ph)

Feature Scope: Inbound only • Lead tracking only (no full conversation transcripts)

Integrations: Facebook Lead Ads, Facebook Messenger, WhatsApp Business API, WordPress (custom plugin)

Document Version: v1.2 (Internal Build Spec)

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1. Executive Summary (For Business & PM)

What problem this solves

Inbound inquiries arrive through multiple channels and get handled inconsistently. This causes:

- Missed or late follow-ups
- Duplicate leads/contacts
- Manual staging and inconsistent pipeline reporting
- Limited visibility on lead source effectiveness

Business value

- **One unified inbound pipeline:** every inbound channel produces a consistent “Lead Event”
- **Fewer duplicates** through automated matching rules
- **Faster and more accurate staging** through AI-assisted classification
- **Cleaner reporting** (source → stage → outcome) because every inbound event is standardized and logged

Expected impact

- Reduced manual data entry (leads created/updated automatically)
- Reduced time from inquiry to correct stage placement
- Improved stage consistency across teams
- Higher conversion due to faster triage and follow-up

Why this is important now

This is foundational for scaling inbound volume without scaling headcount linearly. It enables reliable pipeline reporting and prevents operational breakdown as marketing and inbound traffic increase.

2. User Experience Overview (For UI/UX)

UX Principle: “Tracking Only”

This feature **does not** add a chat inbox or store full message threads. Instead, it stores **Lead Events** (summarized activity + AI signals) that attach to the Lead timeline.

2.1 Admin: Connecting Integrations

Settings → Integrations → Omnichannel Lead Capture

Each channel has a card with:

- Status: Connected / Not connected
- Last inbound event received (timestamp)
- Actions: Connect / Reconnect / Disconnect
- Configuration: Field Mapping, Stage Mapping, AI Settings, Logs

2.2 Screens Added / Updated

A) Integrations Dashboard (New)

Purpose: connect channels, configure mapping, view health.

- Channel cards + status indicators
- “Test connection” action
- “View logs” action

B) Field Mapping (New)

Applies to:

- Facebook Lead Ads
- WordPress forms

UI requirements:

- Left column: external fields
- Right column: CRM fields (standard + custom)

- Validation: required fields, format warnings

C) Stage Mapping + AI Settings (New)

- Map AI “labels” → pipeline stages
- Configure confidence thresholds for:
 - Show suggestion
 - Auto-apply stage (optional)
- Toggle: store message snippet (default OFF)

D) Lead Record → Timeline / Activity Feed (Enhanced)

The lead timeline shows:

- “Lead Event” entries (Messenger inquiry, WhatsApp inquiry, Lead Ad submission, WP form submission)
- “AI Suggestion” entries (stage + confidence + reasons)
- “Manual Override” entries (stage changed by user)

What users see

- A short summary like:
 “Inbound inquiry received via WhatsApp”
 “AI suggested stage: Qualified (82%) – Reasons: asked for pricing, urgent timeline”

What users do NOT see

- No full chat view
- No full transcript
- No message bubble interface

2.3 Workflow Steps (Simple Flow)

1. Inbound event arrives from channel
2. System matches/creates Contact
3. System matches/creates Lead
4. System logs Lead Event
5. AI classifies → suggests stage
6. User accepts or overrides
7. Timeline logs all actions for audit

3. Functional Requirements (Clear & Simple)

A. Facebook Lead Ads

Plain language requirements

- When a Facebook Lead Ad submission is received, the system must:
 - Create or update the Contact
 - Create or update the Lead
 - Record a Lead Event with source = “Facebook Lead Ads”
 - Trigger AI classification
 - Apply stage suggestion rules

Duplicate handling (Contact + Lead)

Match priority:

1. Email
2. Phone (normalized)
3. External lead reference ID (for event idempotency only)

Rules:

- If match found: update missing fields (default)
- Always log a new Lead Event per submission
- Never create duplicate contacts for same email/phone (unless forced)

Field mapping

Admin-configurable mapping:

- Facebook form fields → CRM lead/contact fields
- Support custom fields
- Support “default values” when field is missing

Technical notes

- Use Meta Graph API + Lead Ads Webhooks
- Must implement:
 - Webhook signature verification
 - Idempotency (same lead submission should not create duplicates)
 - Retry-safe processing

B. Facebook Messenger (Lead Tracking Only)

Plain language requirements

- When an inbound Messenger message arrives:
 - Create a **Messenger Lead Event** (not a conversation record)
 - Match/create Contact
 - Match/create Lead

- Trigger AI classification
- Log activity entry in lead timeline

Event grouping rules (tracking only)

- Do not store threads.
- Optionally group multiple events into a single “Recent Messenger Activity” rollup (UI-level grouping), but storage remains event-based.

Contact identification

Primary identifier:

- Page-scoped user identifier (PSID) + Page ID (external reference)

If the lead later submits a form with email/phone:

- Link PSID to the matching contact via verified match rules.

Technical notes

- Use Messenger inbound webhooks
- Store minimal payload:
 - Channel, timestamp, external IDs
 - Optional sanitized snippet (default OFF)
 - AI extracted signals

C. WhatsApp Business API (Lead Tracking Only)

Plain language requirements

- When an inbound WhatsApp message arrives:
 - Create a **WhatsApp Lead Event**
 - Match contact by phone (E.164)
 - Match/create lead
 - Trigger AI classification
 - Log activity in lead timeline

Media support (tracking only)

- Do not store media files by default.
- Store only:
 - has_media = true/false
 - media_type (image/document/audio)
 - optional external media reference ID (for audit/troubleshooting)

Contact matching rules

Primary:

- Phone number normalized to E.164

Conflict handling (same phone on multiple contacts):

- Attach to the most recently active lead/contact
- Flag a “Data conflict” warning in logs for cleanup

Technical notes

- Use WhatsApp Cloud API webhooks
- Validate webhook token/signature
- Normalize phone numbers consistently

D. WordPress Plugin (Custom)

Plain language requirements

- WordPress forms send leads into CRM securely and reliably.
- The plugin must:
 - Authenticate using an API key
 - Push form data via HTTPS
 - Retry transient failures
 - Queue locally if CRM is unreachable

Secure connection

- CRM generates API key
- WP admin configures key in plugin settings
- Requests must be rejected if:
 - key invalid
 - domain restriction mismatch (if enabled)
 - rate limit exceeded

Form submission behavior

Baseline modes:

- Plugin provides its own form block/shortcode; OR
- Plugin hooks into popular form plugins (optional extension)

Error handling

- Retry up to N times (default 3)
- If still failing:
 - store locally as “Pending Sync”

- allow manual resend + automatic scheduled resend

Technical notes

- Endpoint: [POST /api/inbound/wordpress/leads](#)
- Every submission includes:
 - submission_id (UUID) for idempotency
 - timestamp
 - form_id/form_name
 - payload

4. AI Conversation Analysis (Explained Clearly)

Simple explanation

AI does not analyze “full conversation history.”

AI analyzes the **latest inbound lead signal** and structured data such as:

- Form fields (Lead Ads / WordPress)
- The latest message text (if snippet capture is enabled) OR keyword extraction results
- Channel metadata (source, time, presence of media)

Signals AI looks for

- **Intent:** pricing, booking, product inquiry, partnership, support, complaint
- **Urgency:** “now”, “today”, “ASAP”, deadlines, time windows
- **Budget sensitivity:** “how much”, “discount”, “cheapest”, “budget”
- **Fit indicators:** location, category match, requirements
- **Commitment cues:** “schedule a call”, “send proposal”, “ready to proceed”

How AI decides pipeline stage

AI outputs:

- suggested_stage_label (internal AI label)
- confidence_score (0–100)
- reasons/signals (human-readable bullets)

Then system maps the AI label to a real CRM pipeline stage via admin settings.

What confidence score means

- A numeric estimate of certainty.
- Used to:
 - show suggestions more prominently when high

- optionally auto-apply stage when above threshold

When re-analysis happens

- On every new Lead Event arrival
- On new form submission
- Optional manual action: “Re-analyze”

Technical explanation

- Asynchronous processing (does not block lead ingestion)
- Message/event queue for AI jobs
- Triggers:
 - inbound_event_created
 - lead_updated (after mapping)
- Debounce rule:
 - if multiple events arrive quickly, analyze once after short delay (configurable)

5. Data & System Impact (For Developers)

5.1 Data Model (Minimal Storage, Event-Based)

A) Contact (existing object)

Add/ensure fields:

- primary_email
- primary_phone_e164
- facebook_psid (nullable)
- facebook_page_id (nullable)
- whatsapp_phone_e164 (nullable)
- last_inbound_channel
- last_inbound_at

B) Lead (existing object)

Add/ensure fields:

- lead_source (enum: fb_lead_ads, messenger, whatsapp, wordpress)
- pipeline_stage_id
- last_activity_at
- ai_suggested_stage_id (nullable)
- ai_confidence_score (nullable)
- ai_last_classified_at (nullable)

C) Lead Events (new)

Stores tracking-only inbound events (no transcript).

- lead_event_id (PK)
- lead_id (FK)
- contact_id (FK)
- channel (enum)
- external_event_id (nullable)
- external_actor_key (PSID+page_id OR phone OR lead_id)
- received_at
- summary_text (short human-readable, generated)
- snippet_text (nullable; default OFF)
- has_media (bool)
- media_type (nullable)
- payload_min_json (sanitized)

D) AI Classification Results (new)

- ai_result_id (PK)
- lead_event_id (FK)
- suggested_stage_label
- mapped_stage_id (nullable until mapping resolves)
- confidence_score
- reasons_json
- model_version
- created_at

E) Activity Timeline entries (existing or new)

If RIBO already has an activity timeline mechanism, reuse it. Otherwise implement:

- activity_id
- lead_id
- type (lead_event, ai_suggestion, stage_change, error)
- summary
- actor (system/user)
- created_at
- metadata_json

5.2 Inbound Endpoints

- [POST /webhooks/facebook/lead-ads](#)
- [POST /webhooks/facebook/messenger](#)
- [POST /webhooks/whatsapp/messages](#)

- [POST /api/inbound/wordpress/leads](#)

Minimum requirements:

- Auth verification (signature/token/api key)
- Schema validation
- Idempotency handling
- Error classification (4xx vs 5xx) to allow safe retries

5.3 Processing Steps (End-to-End)

1. Receive inbound payload
2. Validate (signature/token/schema)
3. Sanitize + store minimal lead_event
4. Match contact (email/phone/PSID/WA phone)
5. Create/update lead
6. Write timeline entry "Inbound lead event received"
7. Enqueue AI classification job
8. Store AI result + timeline entry
9. If auto-apply enabled and confidence $\geq$ threshold $\rightarrow$ update stage + log "Stage auto-updated"

5.4 Security Validation

- Webhook signature checks (Meta)
- API key management (WordPress)
- Rate limiting per endpoint
- Payload minimization (do not store full message dumps unless explicitly enabled)

5.5 Error Handling

- Retry-safe ingestion (idempotency keys)
- Dead-letter queue for failed AI jobs
- Integration logs viewable by admins only
- Alerts/monitoring for:
 - high webhook failures
 - no inbound events received for X hours
 - queue backlog growth

6. QA Testing Scenarios

Each test includes: **Scenario • Steps • Expected Result**

Positive test cases

TC-01 — Facebook Lead Ad submission creates/updates contact + lead

- Steps:
 - Trigger a test Lead Ad submission
 - Check created/updated Contact
 - Check Lead record and Lead Event
- Expected:
 - Contact matched/created correctly
 - Lead created/updated
 - Lead Event logged with source “Facebook Lead Ads”
 - AI suggestion created

TC-02 — Messenger inbound event creates Lead Event only

- Steps:
 - Send message to FB Page
 - Open lead timeline
- Expected:
 - Lead Event created (no conversation thread stored)
 - Contact matched/created
 - AI suggestion created

TC-03 — WhatsApp inbound event with media

- Steps:
 - Send WA message with image
 - Review lead timeline
- Expected:
 - Lead Event shows “has_media=true, media_type=image”
 - No media file stored by default
 - AI suggestion created

TC-04 — WordPress form submission succeeds

- Steps:
 - Submit form from WP
 - Verify lead/contact created/updated
- Expected:
 - Event logged
 - Fields mapped properly
 - AI suggestion created

Negative test cases

TC-05 — Invalid webhook signature

- Steps:
 - Send a webhook request with invalid signature
- Expected:
 - Rejected (401/403)
 - No Lead Event created
 - Security log entry created

TC-06 — Malformed payload

- Steps:
 - Send inbound payload missing required fields
- Expected:
 - Rejected (400)
 - Error logged

Duplicate lead scenarios

TC-07 — Same email submits FB Lead Ad twice

- Steps:
 - Submit twice using same email
- Expected:
 - Single Contact record
 - Lead updated
 - Two Lead Events logged
 - No duplicates created

TC-08 — WhatsApp message from existing phone

- Steps:
 - Create Contact with phone
 - Send WA message from same phone
- Expected:
 - Contact matched, not duplicated
 - Lead Event attaches to same lead

AI misclassification scenario

TC-09 — AI suggests wrong stage, user overrides

- Steps:
 - Generate inbound event with ambiguous intent
 - AI suggests stage
 - User manually changes stage
- Expected:

- Manual stage overrides AI
- Timeline logs override with user identity
- AI suggestion remains recorded

Webhook failure scenario

TC-10 — Provider retries after 500

- Steps:
 - Simulate ingestion endpoint returns 500
 - Provider retries same event
- Expected:
 - Only one Lead Event created (idempotency)
 - Logs show retry + eventual success

Plugin authentication failure scenario

TC-11 — WordPress plugin invalid API key

- Steps:
 - Set invalid key in plugin
 - Submit a form
- Expected:
 - CRM returns 401
 - Plugin logs failure
 - Submission queued locally (Pending Sync), if enabled

7. Admin Configuration & Controls

Integration setup dashboard

- Connect/disconnect each channel
- Last event timestamp
- Error counters (24h/7d)
- Test connection button
- Logs viewer

API key management (WordPress)

- Generate key
- Rotate key
- Disable key
- Optional restrictions:
 - Allowed domain

- IP allowlist

Stage mapping rules

- AI labels → CRM pipeline stages mapping
- Per-channel overrides (optional)

AI settings

- Enable AI suggestions (on/off)
- Confidence threshold:
 - Show suggestion threshold (e.g., ≥ 40)
 - Auto-apply threshold (e.g., ≥ 85)
- Snippet capture toggle (default OFF)

Enable/disable toggles

- Per integration and per sub-feature (AI, auto-apply, snippet capture)

8. Non-Functional Requirements (Simplified)

Performance

- Webhook ingestion responds within seconds
- AI classification asynchronous (non-blocking)
- System handles burst traffic (campaign spikes)

Security

- Signature/token validation for Meta webhooks
- HTTPS only
- Encryption for tokens/API keys at rest
- Least-privilege access control for integration settings/logs

Scalability

- Queue-driven AI processing
- Worker autoscaling (or manual scaling plan)

Data privacy

- Minimal event storage
- No transcripts by default
- Configurable retention policy for lead events

Logging & monitoring

- Integration health dashboard
- Error logs with correlation IDs
- Alerts for:
 - webhook failure spikes
 - no inbound events in X hours
 - AI queue backlog

9. Risks & Dependencies

API dependency risks

- Meta API outages, throttling, schema changes
- Webhook delivery delays/retries

Policy limitations

- WhatsApp Business verification and compliance requirements
- Facebook platform policy changes affecting data access

AI accuracy risks

- Misclassification causing wrong stage routing if auto-apply is too aggressive
- Requires threshold tuning and monitoring

Data privacy risks

- Inbound messages may contain sensitive personal data
- Mitigation: minimal storage, snippet OFF, retention controls, access control

10. Success Metrics (KPIs)

AI metrics

- **Suggestion acceptance rate (%)**
Accepted AI stage suggestions / total suggestions
- **Stage accuracy (sampled %)**
QA-reviewed correctness rate (weekly sample)

Operational metrics

- **Time-to-stage (median minutes)**
Time from inbound event to lead placed in correct stage
- **Manual data entry reduction (%)**
Reduction in manually created leads/contacts

Business metrics

- **Qualified leads increase (%)**
Qualified leads per channel compared to baseline
- **Conversion improvement (%)**
Inquiry → Qualified → Closed rate change

Implementation Notes for RIBO Team (Internal Alignment)

Out of scope (explicit)

- Full inbox / conversation UI
- Full transcript storage and retrieval
- Outbound messaging/replies through Messenger/WhatsApp
- Media file storage (beyond “received” flags)

Core deliverable

Reliable, secure ingestion of inbound lead signals + dedupe + standardized lead events + AI stage suggestion + audit logs + admin controls.